

AI-Powered IT Incident Prediction & Prevention System

1K

Total Records

98

Total Incidents

0.10

Incident Rate %

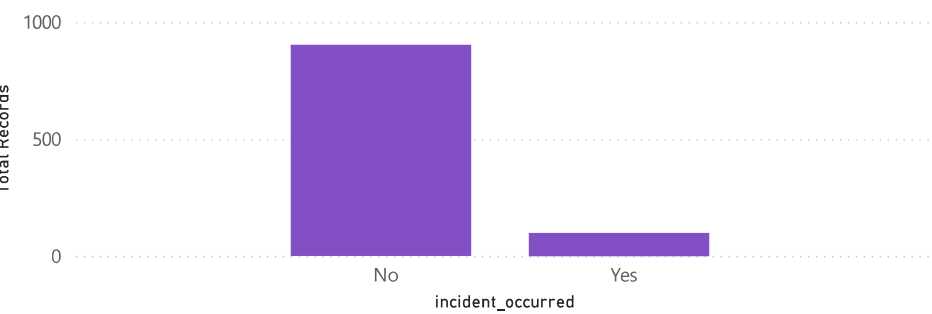
5

Open_Incidents

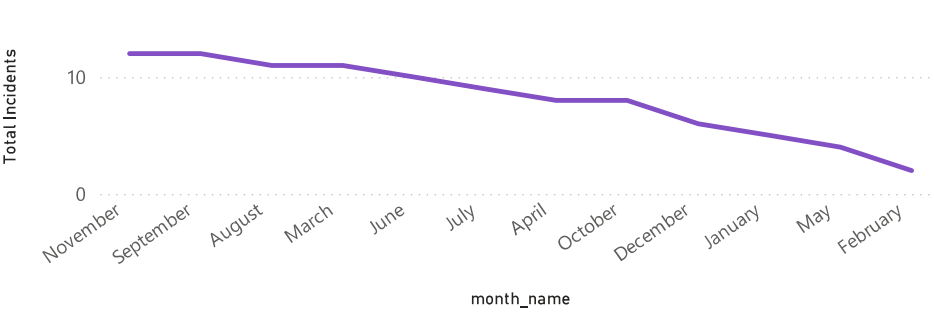
10.95

Average Anomaly Score

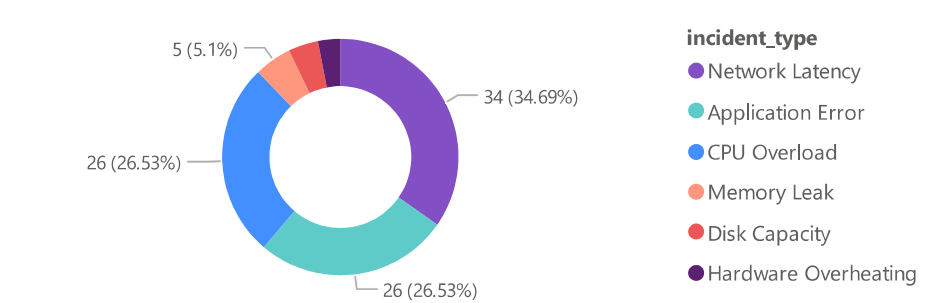
Total Records by incident_occurred



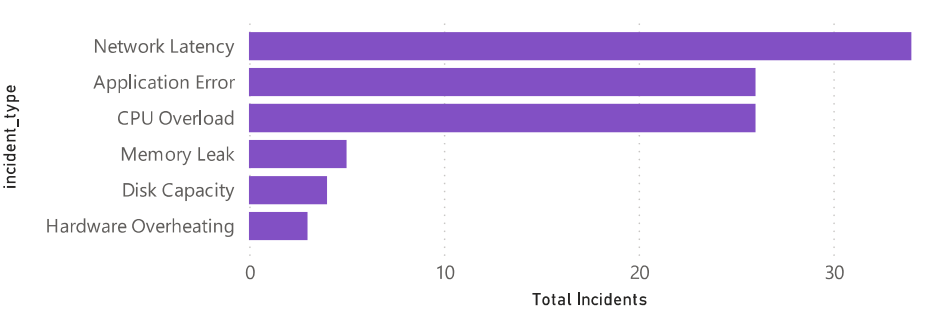
Total Incidents by month_name



Total Incidents by incident_type



Total Incidents by incident_type



month_name

All

department

All

environment

All

severity

All

System Performance Analysis

CPU Usage

0.53

Memory Usage

0.54

Disk Usage

0.63

Response Time

392.77

Network Latency

60.09

Temperature

56.26

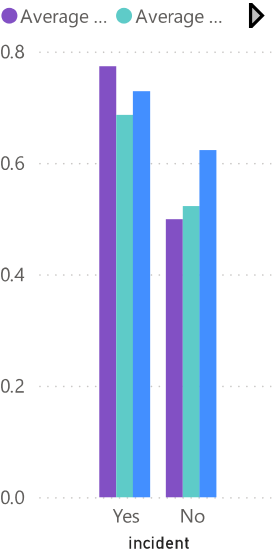
Anomaly Score

10.95

Error Count

3.46

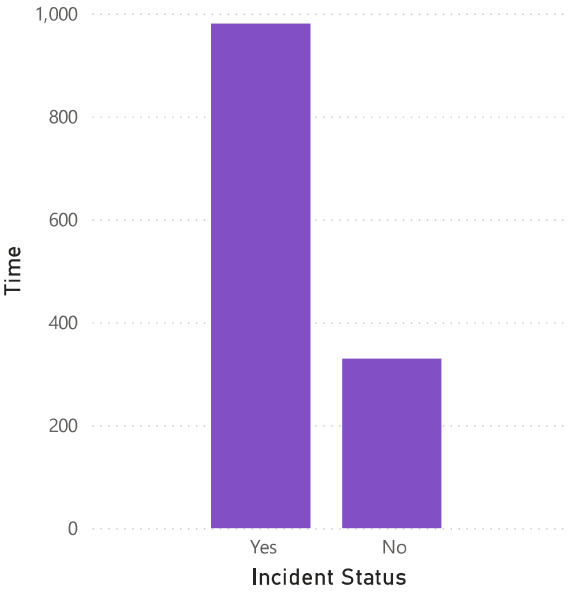
CPU vs Disk vs incident



department

All

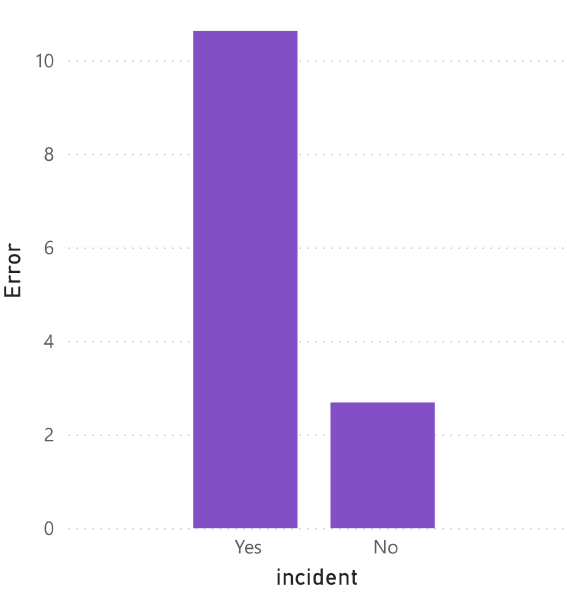
Response Time: Normal vs Incident



environment

All

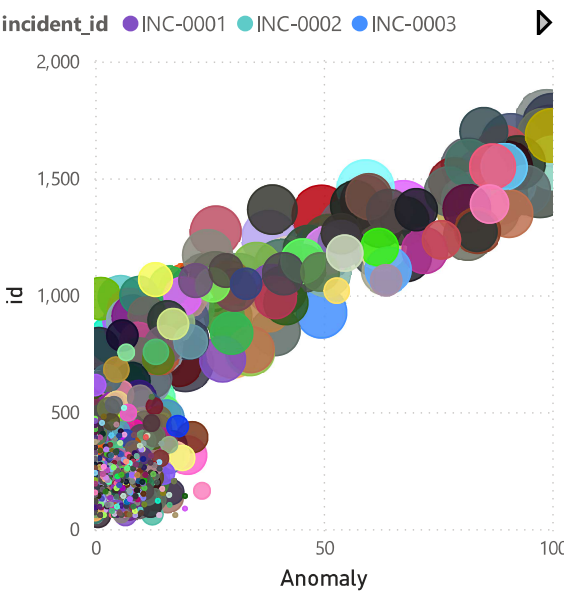
Error vs incident



incident_occurred

All

anomaly vs Time vs Error vs incident vs id



month_name

All

Incident Analysis

28

Critical Incidents

78

Resolved Incidents

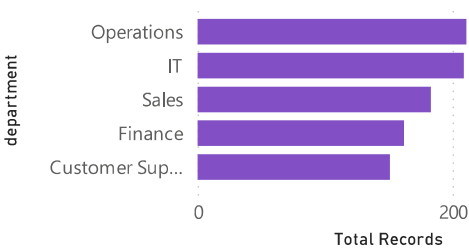
15

Monitoring Incidents

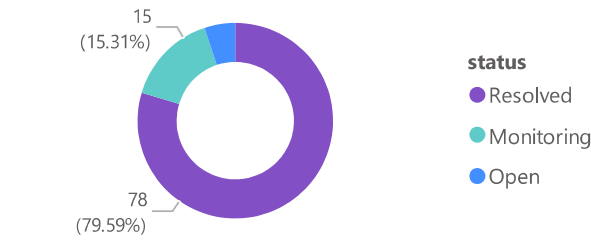
5

Open Incidents

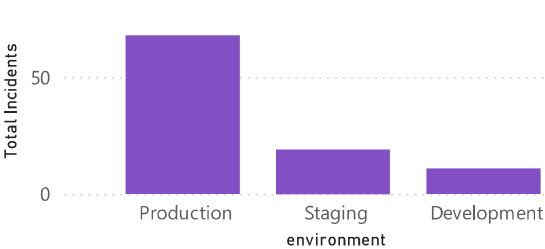
Total Records by department



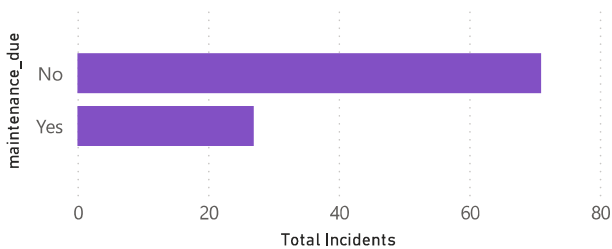
Total Incidents by status



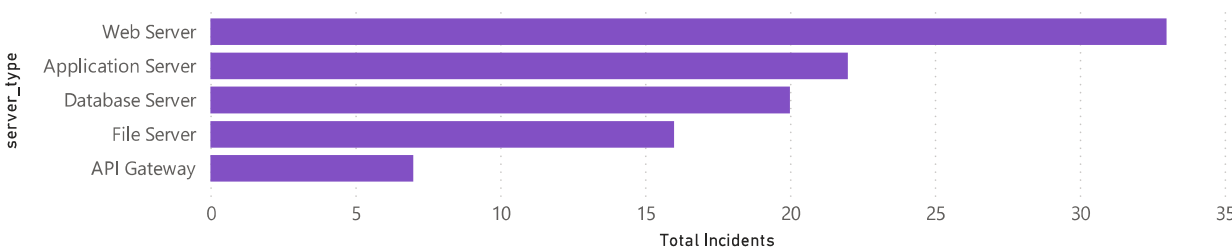
Total Incidents by environment



Total Incidents by maintenance_due



Total Incidents by server_type



server_id	department	environment	server_type	Sum of cpu_
SRV-001	Customer Support	Development	Web Server	
SRV-001	Customer Support	Production	Application Server	
SRV-001	Customer Support	Production	File Server	
SRV-001	Customer Support	Production	Web Server	
SRV-001	Customer Support	Production	Web Server	
SRV-001	Customer Support	Staging	Web Server	
SRV-002	Customer Support	Production	API Gateway	
SRV-002	Customer Support	Production	Application Server	
SRV-002	Customer Support	Production	Web Server	
SRV-003	Customer Support	Staging	Application Server	
SRV-004	Customer Support	Production	Web Server	
SRV-005	Customer Support	Production	Application Server	
SRV-005	Customer Support	Production	File Server	
SRV-005	Customer Support	Staging	API Gateway	
SRV-006	Customer Support	Production	Application Server	
SRV-006	Customer Support	Production	Web Server	
SRV-007	Customer Support	Development	Application Server	
SRV-007	Customer Support	Production	Database Server	
SRV-007	Customer Support	Production	File Server	
SRV-007	Customer Support	Staging	Web Server	
SRV-008	Customer Support	Development	Application Server	
SRV-008	Customer Support	Production	Application Server	
SRV-008	Customer Support	Production	Application Server	
SRV-008	Customer Support	Production	Web Server	

Total

Project Summary & Business Insights

1K	98	0.10	28	5
Total Records	Total Incidents	Incident Rate %	Critical Incidents	Open_Incidents

Key Findings

- The dataset contains 1,000 IT system monitoring records.
- 98 IT incidents were detected.
- The incident rate is approximately 9.80%.
- High CPU usage, response time, error count, network latency and anomaly score are important risk indicators.
- Production and high-usage systems require stronger monitoring.
- Open and critical incidents need faster resolution and preventive action.

Business Recommendations

- Monitor high-risk servers continuously.
- Set automatic alerts for high anomaly scores.
- Prioritize critical and open incidents.
- Schedule preventive maintenance for risky servers.
- Use CPU, memory, disk, response time and error count as early-warning indicators.
- Review departments and server types with high incident rates.

Final Conclusion

The AI-Powered IT Incident Prediction & Prevention System helps IT teams understand system behavior, identify high-risk infrastructure and take proactive action before failures become major outages.

The Power BI dashboard provides clear visibility into incident trends, severity, server performance, environment risk, maintenance status and operational reliability.

This project supports faster decision-making, reduced downtime and